Experiment 2.1.1

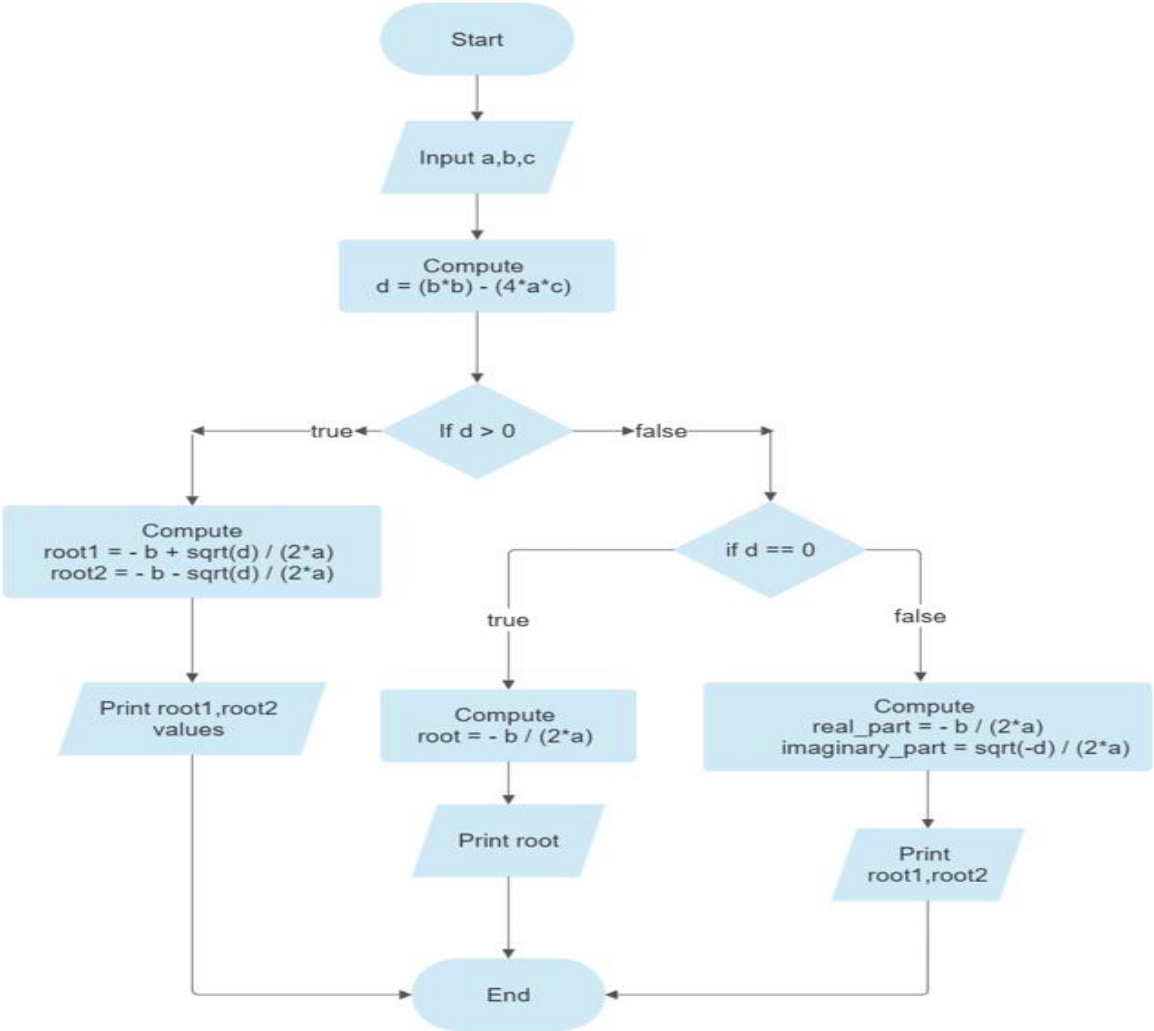
Algorithm to Find Roots of a Quadratic Equation

Equation: $ax^2 + bx + c = 0$

Step 1: Start
Step 2: Read coefficients a, b, c
Step 3: Compute discriminant ($D = b^2 - 4ac$)

Step 4:

- If D > 0
  - root1 = $(-b + \sqrt{D}) / (2a)$
  - root2 = $(-b - \sqrt{D}) / (2a)$
- Else if D = 0
  - root = -b / (2a)
- Else
  - real part = -b / (2a)
  - imaginary part = $\sqrt{D}$ / (2a)
  - **Step 5:** Display roots
    Step 6: Stop



2.1.1. Roots of a Quadratic Equation

55:17

Write a program to find the roots of a quadratic equation, given its coefficients a, b, and c. Use the quadratic formula:

$$\frac{\left(-b \pm \sqrt{b^2 - 4ac}\right)}{2a}$$

The discriminant $D = b^2 - 4ac$ determines the nature of the roots:

- If $D > 0$: Roots are real and different
- If $D = 0$: Roots are real and the same
- If $D < 0$: Roots are imaginary

Input Format:

- Three space-separated integers representing the coefficients a, b, and c, respectively.

Output Format:

- If roots are real and different, print:

```
root1 = <Root1>
root2 = <Root2>
```

- If roots are the same, print:

```
root1 = root2 = <Root1>
```

- If roots are imaginary, print:

```
root1 = <RealPart>+<ImaginaryPart>i
root2 = <RealPart>-<ImaginaryPart>i
```

- All values should be formatted to two decimal places.

Sample Test Cases +

Explorer

quadratic... Submit

```python
import math
a, b, c = map(int, input().split())
d = b*b - 4*a*c

if d > 0:
    r1 = (-b + math.sqrt(d)) / (2*a)
    r2 = (-b - math.sqrt(d)) / (2*a)
    print(f"root1 = {r1:.2f}")
    print(f"root2 = {r2:.2f}")

elif d == 0:
    r = -b / (2*a)
    print(f"root1 = root2 = {r:.2f}")
else:
    real = -b / (2*a)
    imag = math.sqrt(-d) / (2*a)
    print(f"root1 = {real:.2f}+{imag:.2f}i")
    print(f"root2 = {real:.2f}-{imag:.2f}i")
```

Debugger

Average time	Maximum time
0.009 s	0.017 s
9.33 ms	17.00 ms

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 17 ms Debug

Expected output	Actual output
1 -5 6	1 -5 6
root1 = 3.00	root1 = 3.00
root2 = 2.00	root2 = 2.00

Test case 2 7 ms

Terminal Test cases